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Gas generator

Abstract

To obtain a gas generator that is easier to manufacture than before. A gas generator **100** has a long substantially columnar outer shape, and includes a housing **10**, a first holder **20**, a second holder **25**, and a third holder **53** attached near one opening end of the housing **10**, an igniter **50** that is held by the second holder **25** and ignites a gas generating agent **31**, a closing member **12** attached to the other end of the housing **10** so as to close the other opening end of the housing **10**, and a filter **41**. The second holder **25** is fixed to the housing **10** by forming a circumferential wall **10d** and a circumferential wall **10e** in a state where the second holder **25** is sandwiched between the first holder **20** and the third holder **53**. The third holder **53** is fixed to the housing **10** by performing diameter reduction processing (caulking processing or the like) from an outer circumference to an inner side of the housing **10** at a position corresponding to an annular groove portion **54** to form the circumferential wall **10d**.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a gas generator incorporated in an airbag device as an occupant protection device mounted on an automobile or the like, and more particularly to a so-called cylinder-type gas generator having a long cylindrical shape.

BACKGROUND ART

(2) A long cylindrical housing in a cylinder-type gas generator is generally configured such that one end thereof is closed by a closing member and the other end thereof is closed by a holder having an ignition portion (See, for example, Patent Literature 1 below).

CITATIONS LIST

Patent Literature

(3) Patent Literature 1: JP 6579992 B2

SUMMARY OF INVENTION

Technical Problems

(4) However, since the outer shape of the holder in the gas generator of the above Patent Literature is a complicated shape, the manufacturing becomes complicated even if the holder is cut or cast using a mold.

(5) Therefore, an object of the present invention is to provide a gas generator that is easier to manufacture than before.

Solutions To Problems

(6) (1) A gas generator includes: a long cylindrical housing that is loaded with a gas generating agent that generates a gas by burning, includes a filter inside through which the gas passes, and has a gas ejection port that ejects the gas formed at a position corresponding to the filter; an igniter capable of igniting and combusting the gas generating agent; a tubular holder that holds a part of the igniter and is fixed to one end portion of the housing in an axial direction; and a closing member fixed to another end portion of the housing in the axial direction, in which the holder includes: a tubular first holder disposed on a side of the one end portion of the housing and capable of fitting an external connector that energizes the igniter on the side of the one end portion; a tubular second holder disposed on an inner side of the housing of the first holder in a state of being in contact with the first holder, in which an outer diameter of at least a portion (hereinafter, a contact portion) in contact with an inner wall of the housing is larger than an outer diameter of the first holder on a side of the other end portion or a diameter same as an outer diameter of the first holder on the side of the other end portion; and a tubular third holder including, on an outer side surface, at least a portion (hereinafter, a small-diameter portion) having a diameter smaller than a diameter of an end portion on a side opposite to a side of the second holder, and disposed on an inner side of the housing with respect to the second holder in a state of being in contact with the second holder, and a portion of the contact portion of the second holder corresponding to a portion on a side of the first holder and a portion corresponding to the small-diameter portion of the third holder are subjected to diameter reduction processing from an outer side of the housing, and the holder is fixed to the housing by sandwiching at least a portion of the first holder in contact with the second holder and a portion of the third holder in contact with the second holder together with the second holder with a portion of the housing subjected to the diameter reduction processing. (2) In the gas generator of (1) described above, it is preferable that at least a part on a side of one end portion of the second holder is fitted and inserted into the first holder, and another end portion of the first holder is fixed to the housing in a state of being sandwiched between the inner wall of the housing of a portion subjected to the diameter reduction processing at a location corresponding to an annular groove portion of the first holder and at least a part on a side of one end portion of the

holder. (3) In the gas generator of (1) or (2) described above, it is preferable that a brim portion is provided on a side of the closing member of the third holder, and at least at the time of activation, a tip portion of the brim portion is in contact with the inner wall of the housing. (4) As another aspect, a gas generator of the present invention includes: a long cylindrical housing that is loaded with a gas generating agent that generates a gas by burning, includes a filter inside through which the gas passes, and has a gas ejection port that ejects the gas formed at a position corresponding to the filter: an igniter capable of igniting and combusting the gas generating agent and including a tubular cup-shaped member on a side of the gas generating agent; a tubular holder that holds a part of the igniter and is fixed to one end portion of the housing in an axial direction; and a closing member fixed to another end portion of the housing in the axial direction, in which the holder includes: a tubular first holder disposed on a side of the one end portion of the housing and capable of fitting an external connector that energizes the igniter on the side of the one end portion: a tubular second holder disposed on an inner side of the housing of the first holder in a state of being in contact with the first holder, in which an outer diameter of at least a portion (hereinafter, a contact portion) in contact with an inner wall of the housing is larger than an outer diameter of the first holder on a side of the other end portion or a diameter same as an outer diameter of the first holder on the side of the other end portion: and a tubular third holder including, on an outer side surface, at least a portion (hereinafter, a small-diameter portion) having a diameter smaller than a diameter of an end portion on a side opposite to a side of the second holder, a brim portion being provided on a side of the closing member, the third holder being disposed on an inner side of the housing with respect to the second holder such that a tip portion of the brim portion comes into contact with the inner wall of the housing in a state of being in contact with the second holder and at least at the time of activation, a portion of the contact portion of the second holder corresponding to a portion on a side of the first holder and a portion corresponding to the small-diameter portion of the third holder are subjected to diameter reduction processing from an outer side of the housing, and the first holder and the second holder are fixed to the housing by sandwiching at least a portion of the first holder in contact with the second holder together with the second holder with a portion of the housing subjected to the diameter reduction processing, and the third holder is fixed to the igniter by press-fitting an inside of the third holder into the cup-shaped member of the igniter. (5) In the gas generator of (1) to (4) described above, it is preferable that a surface of the third holder on a side of the gas generating agent is formed such that the surface of the third holder on the side of the gas generating agent formed around a tip portion of the igniter is flush with a surface of the tip portion of the igniter. (6) In the gas generator of (1) to (5) described above, it is preferable that the second holder is a tubular member including a fitting portion that fits and holds the igniter inside, and a caulking portion that is provided at an opening end portion on a side of the other end portion and caulks and fixes the igniter, and the third holder is formed to cover the caulking portion. (7) In the gas generator (1) to (6) described above, it is preferable that the first holder and the third holder are made of resin, and the second holder is made of metal or alloy.

Advantageous Effects of Invention

(7) According to the present invention, it is possible to provide a gas generator that is easier to manufacture than before.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic view (partially omitted) illustrating an internal structure of a gas generator according to a first embodiment of the present invention partially illustrated in cross section.

(2) FIG. 2 is a schematic view (partially omitted) illustrating an internal structure of a gas generator according to a second embodiment of the present invention partially illustrated in cross section.

(3) FIG. 3 is a schematic view (partially omitted) illustrating an internal structure of a gas generator according to a third embodiment of the present invention partially illustrated in cross section.

(4) FIG. 4 is a schematic view (partially omitted) illustrating an internal structure of a gas generator according to a fourth embodiment of the present invention partially illustrated in cross section.

DESCRIPTION OF EMBODIMENTS

First Embodiment

(5) Hereinafter, an internal structure of a cylinder-type gas generator according to an embodiment of the present invention will be described with reference to FIG. 1.

Configuration of Gas Generator 100

(6) A gas generator **100** has a long substantially columnar outer shape, and includes a housing **10**, a first holder **20**, a second holder **25**, and a third holder **53** attached near one opening end of the housing **10**, and a closing member **12** attached to the other end of the housing **10** so as to close the other opening end of the housing **10**. Note that the first holder **20**, the second holder **25**, and the third holder **53** constitute a holder that holds an igniter **50**.

(7) The housing **10** includes a long cylindrical member having circumferential walls **10a**, **10b**, **10c**, **10d**, and **10e** and having openings at both ends in an axial direction. The closing member **12** includes annular groove portions **13** and **14** formed by reducing a diameter of a circumferential surface of a bottomed tubular member having a predetermined thickness by caulking (an example of a diameter reduction processing method) and fixing described later, and is formed in a substantially vessel shape including an internal space **12a**. The annular groove portions **13** and **14** formed by the caulking and fixing are formed on the circumferential surface of the closing member **12** so as to extend in a circumferential direction. Furthermore, a gas ejection port **11** is provided in a circumferential wall in the vicinity of an end portion of the housing **10** on a side where the closing member **12** is attached. The gas ejection port **11** is a hole for ejecting a gas generated inside the gas generator **100** to the outside, and a plurality of the gas ejection ports **11** are provided along the circumferential direction and the axial direction of the housing **10**.

(8) Furthermore, the closing member **12** is made of metal such as stainless steel, steel, aluminum alloy, or stainless alloy. Then, as illustrated in FIG. 1, in a state where a part of the closing member **12** is inserted into one opening end of the housing **10**, the circumferential walls **10a** and **10b** of the housing **10** at a portion corresponding to a part of the circumferential surface of the closing member **12** are radially reduced (caulked) inward to form the annular groove portions **13** and **14**, whereby the closing member **12** is caulked and fixed to the housing **10**.

(9) The first holder **20** is a substantially tubular member made of resin, and includes a fitting portion **21** to which a female connector (not illustrated) for energizing the igniter **50** can be fitted on an opposite side of a holding position of the igniter **50**, an annular groove portion **22** formed on an outer circumferential surface so as to extend in the circumferential direction, and a flange portion **26** formed at an end portion of the annular groove portion **22** on a side of the second holder **25**. The flange portion **26** is formed to have an outer diameter smaller than an outer diameter of an annular portion **27** of the second holder **25**. The annular portion **27** includes, on an outer circumference, a portion (contact portion) in contact with the inner wall of the housing. Note that the circumferential wall **10e** of the housing **10** in a portion corresponding to the flange portion **26** provided on the outer circumferential surface of the first holder **20** is radially reduced (caulked) inward to engage with the flange portion **26**, whereby the first holder **20** is caulked and fixed to the housing **10**. Furthermore, an end portion of the second holder **25** on a side of the first holder **20** is inserted into the first holder **20**. As a result, in a case where a diameter of the circumferential wall **10e** of the housing **10** is reduced radially inward, the flange portion **26** is sandwiched between the end portion of the second holder **25** on the side of the first holder **20** and an inner wall of the circumferential wall **10e** of the housing **10**, and the first holder **20** can be more firmly fixed to the housing **10**. That is, sealability between the housing **10** and the holder can be secured. Here, as one modification, instead of the flange portion **26**, as a simple end portion (having no flange protruding

in a radial direction) continuous from a surface of the annular groove portion **22**, the end portion may be sandwiched between the end portion of the second holder **25** on the side of the first holder **20** and the inner wall of the circumferential wall **10e** of the housing **10**.

(10) Here, as a raw material of the first holder **20**, a resin material excellent in heat resistance, durability, corrosion resistance, and the like after curing is suitably selected and used. In this case, the raw material is not limited to a thermosetting resin typified by an epoxy resin or the like, a thermoplastic resin typified by a polybutylene terephthalate resin, a polyethylene terephthalate resin, a polyamide resin (for example, nylon 6, nylon 66, or the like), a polypropylene sulfide resin, a polypropylene oxide resin, a polyacetal resin, or the like can also be used. In a case where these thermoplastic resins are selected as raw materials, it is preferable to contain glass fibers or the like as fillers in these resin materials in order to secure the mechanical strength of the first holder **20** after molding. However, in a case where sufficient mechanical strength can be secured only by the thermoplastic resin, it is not necessary to add the filler as described above. The third holder **53** can also be formed of the same raw material as the first holder **20**.

(11) Note that, as described above, the female connector is formed in the fitting portion **21** of the first holder **20**. The female connector is a portion to which a male connector of a harness that transmits a signal from a collision detection means provided separately from the gas generator **100** is connected. A retainer (not illustrated) is attached to the female connector. This retainer is attached to prevent the cylinder-type gas generator **100** from malfunctioning due to electrostatic discharge or the like at the time of conveying the gas generator **100** or the like, and when the male connector of the harness is inserted into the female connector at the assembling stage to an airbag device, the contact with a terminal pin **52** is released.

(12) A portion of the second holder **25** in contact with the inner wall of the housing **10** is larger than an outer diameter of the flange portion **26** of the first holder **20**, and is a substantially tubular member made of metal such as stainless steel or steel, an aluminum alloy, a stainless alloy, or the like. Furthermore, the second holder **25** includes a tapered fitting portion **23** into which a part (mainly a main body portion) of the igniter **50** is inserted and into which the igniter **50** is fitted, and a part of the outside is fitted and inserted into the first holder **20**. Note that the igniter **50** is caulked and fixed to the second holder **25** by a caulking portion **24** provided at the opening end portion. Furthermore, the second holder **25** is fixed to the housing **10** by forming a circumferential wall **10d** described later and the circumferential wall **10e** described above in a state where the second holder **25** is sandwiched between the first holder **20** and the third holder **53**.

(13) The third holder **53** is a substantially tubular member made of resin, and has an annular groove portion **54** (small-diameter portion) for caulking and fixing formed on the outer circumferential surface so as to extend in the circumferential direction, and a surface **55** that is flush or substantially flush with the surface of the tip portion of a squib cup **51**. The third holder **53** is fixed to the housing **10** by performing diameter reduction processing (caulking processing or the like) from the outer circumference to the inner side of the housing **10** at a position corresponding to the annular groove portion **54** to form the circumferential wall **10d**. Furthermore, a part of the igniter **50** (mainly the squib cup **51**) and the caulking portion **24** are inserted into the third holder **53**. Furthermore, the third holder **53** is also a directional member that directs a direction of the flame generated in the igniter **50** toward a gas generating agent **31** at the time of activation. Here, as one modification, the annular groove portion **54** does not necessarily have a groove shape, and may be formed so as to include at least a portion having a diameter smaller than a diameter of the end portion (portion near the surface **55**) on a side (side of the gas generating agent **31**) opposite to the side of the second holder **25** of the third holder **53** to such an extent that the annular groove portion **54** can be used for fixing by diameter reduction processing.

(14) As illustrated in FIG. **1**, the igniter **50** as an ignition means of the gas generating agent **31** is disposed at one end portion (that is, a portion close to the holder (first holder **20**, second holder **25**, and third holder **53**)) of the housing **10** in the axial direction. Note that the holder (first holder **20**,

second holder **25**, and third holder **53**) that fixes the igniter **50** and the igniter **50** has a function as an ignition means that generates flame for burning the granular gas generating agent **31** to be described later.

(15) As illustrated in FIG. **1**, the igniter **50** is held together with the third holder **53** described later in a state of being inserted into the fitting portion **23** of the second holder **25**. More specifically, the igniter **50** includes a base frame through which a pair of the terminal pins **52** is inserted and held, and the squib cup **51** (cup-shaped member) attached on the base frame, and a resistor (bridge wire) is attached so as to connect a tip of the terminal pin **52** inserted into the squib cup **51**, and the squib cup **51** is filled with an igniter powder so as to surround the resistor or to be in contact with the resistor. As the resistor, a nichrome wire or the like is generally used, and as the igniter powder, zirconium potassium perchlorate (ZPP), zirconium tungsten potassium perchlorate (ZWPP), lead trisulfate, or the like is generally used. Note that the inside of the squib cup **51** may be further filled with not only the igniter powder but also an enhance agent, and as the enhance agent that can be disposed simultaneously with the igniter powder, a composition including a metal/oxidizing agent represented by boron/potassium nitrate or the like, a composition including titanium hydride/potassium perchlorate, a composition including boron/5-aminotetrazole/potassium nitrate/molybdenum trioxide, or the like is used.

(16) When the collision is detected, a predetermined amount of current flows through the resistor via the terminal pin **52**. When the predetermined amount of current flows through the resistor, Joule heat is generated in the resistor, and the igniter powder starts combustion by receiving this heat. The hot flame resulting from the combustion ruptures the squib cup **51** containing the igniter powder. The time from when the current flows through the resistor to when the igniter **50** activates is 2 milliseconds or less in a case where a nichrome wire is used as the resistor.

(17) Furthermore, the squib cup **51** is generally made of metal or resin. Note that the circumferential wall portion of the squib cup **51** other than the vicinity of the tip portion is caulked and fixed to the second holder **25** by the caulking portion **24** together with the igniter **50**.

(18) In the housing **10**, the gas generating agent **31**, an AI agent **32**, a winding spring **35**, and a bypass preventing member **36** are disposed in a space between the igniter **50** and the filter **41**.

(19) The gas generating agent **31** is an integrally molded product that is ignited by flame generated by being ignited by the igniter **50** and generates gas by burning. Furthermore, the gas generating agent **31** is generally formed as a molded body containing fuel, an oxidant, and an additive. As the fuel, for example, a triazole derivative, a tetrazole derivative, a guanidine derivative, an azodicarbonamide derivative, a hydrazine derivative, or the like, or a combination thereof is used. Specifically, for example, nitroguanidine, guanidine nitrate, cyanoguanidine, 5-aminotetrazole, or the like is suitably used. Furthermore, examples of the oxidizing agent include basic metal nitrates such as basic copper nitrate, basic metal carbonates such as basic copper carbonate, perchlorates such as ammonium perchlorate or potassium perchlorate, and nitrates containing cations selected from alkali metals, alkaline earth metals, transition metals, and ammonia. As the nitrate, for example, sodium nitrate, potassium nitrate or the like is suitably used. Furthermore, examples of the additive include a binder, a slag forming agent, and a combustion adjusting agent. As the binder, for example, a cellulose derivative such as hydroxypropylene methyl cellulose, an organic binder such as a metal salt of carboxymethyl cellulose or a stearate, and an inorganic binder such as synthetic hydroxytalcite or acidic clay can be suitably used. Silicon nitride, silica, acid clay, and the like can be suitably used as the slag forming agent. Furthermore, as the combustion adjusting agent, metal oxide, ferrosilicon, activated carbon, graphite, and the like can be suitably used.

(20) The bypass preventing member **36** has a bottom surface portion **36a** having a weak member at a central portion and an outer circumferential wall portion **36b**, and is welded and fixed to the housing **10** via a circumferential wall **10c** which is also a welded portion. The bypass preventing member **36** is also a member that partitions the housing **10** into the filter **41** and the combustion chamber of the gas generating agent **31**, and the weak member of the bottom surface portion **36a** is

broken by the pressure of the generated gas at the time of activation. Furthermore, the bypass preventing member **36** is disposed such that the bottom surface portion **36a** is in contact with the end portion of the filter **41** and the outer circumferential wall portion **36b** is in contact with the inner wall of the housing **10**. As a result, the generated gas can be bypassed from between the inner wall of the housing **10** and the outer circumferential portion of the filter **41**, and can be prevented from leaking to the gas ejection port **11**. That is, the bypass preventing member **36** can cause the gas generated at the time of activation to flow into a side of the filter **41** through the ruptured central portion.

(21) The winding spring **35** is disposed such that one end thereof is in contact with a tip side (flame injection side) of the igniter **50**. Furthermore, the winding spring **35** is formed in a spiral shape whose diameter increases from one end side to the other end side, and presses the gas generating agent **31** against a side of the bypass preventing member **36** at the other end side.

(22) The AI agent **32** is provided in a substantially central portion of the bottom surface portion **36a** of the bypass preventing member **36** on the side of the igniter **50**, and has an auto-ignition (AI) function of automatically igniting without depending on the activation of the igniter **50**. More specifically, since the AI agent **32** automatically ignites at a temperature lower than that of the gas generating agent **31**, it is possible to prevent induction of an abnormal operation of the gas generator **100** due to heating from the outside in a case where a fire or the like occurs in a vehicle or the like equipped with an airbag device or the like in which the gas generator **100** is incorporated.

(23) The filter **41** is formed of a cylindrical member having a substantially cylindrical hollow portion **41a** at the center. Note that, by using the filter **41** made of the above-described cylindrical member, the flow resistance of a working gas flowing at the time of activation is suppressed to be low, and efficient gas flow is possible. As the filter **41**, for example, a wire material made of metal such as stainless steel or steel, a wire material obtained by winding a net material, a wire material obtained by pressing a net material, or the like is used. Specifically, a knitted wire mesh, a plain weave wire mesh, an aggregate of crimp weave metal wire materials, or the like is used. When the gas generated in the housing **10** passes through the filter **41**, the filter **41** functions as a cooling means for cooling the gas by depriving high temperature heat of the gas, and also functions as a removal means for removing slag and the like contained in the gas. Here, as a modification of the filter **41**, a filter having a labyrinthine flow path formed by combining substantially cylindrical or mortar parts made of metal may be used. As a result, since the course of the working gas can be changed in various directions, it is possible to cool the gas and remove the slag.

(24) Furthermore, in the embodiment of the present invention described above, the case of using what is called knitted wire mesh as the filter has been exemplified, but instead of this, it is also possible to use a filter manufactured by winding a punching metal or a filter manufactured by winding an expanded metal. Here, the punching metal is a metal plate in which only an opening portion is provided in the plate-shaped metal member (that is, a protrusion portion is not provided on a circumferential edge of the opening portion), and the expanded metal is a metal plate in which openings are provided in the plate-shaped metal member to form a mesh shape by cutting the plate-shaped metal member in, for example, a staggered manner and expanding the cuts. Even in a case where such punching metal or expanded metal is used instead of the above-described knitted wire mesh, the same effects as the effects described in the embodiment of the present invention can be obtained.

(25) Furthermore, in the punching metal and the expanded metal described above, a filter including a laminate is configured by winding a single metal plate-like member, but the configuration of the filter is not limited to this configuration. That is, each layer may be formed of separate metal plate-like members, and the layers may be combined to form a filter formed of a laminate, or a part of a plurality of layers may be formed by winding a single metal plate-like member, and the remaining layers may be formed by winding another single metal plate-like member, and the layers may be

combined to form a filter formed of a laminate.

(26) Next, operation of the gas generator **100** described above at the time of activation will be described. In a case where the vehicle mounted with the airbag device in which the gas generator **100** in the present embodiment is incorporated collides, the collision is detected by a collision detection means separately provided in the vehicle, and the igniter **50** is operated based on the collision. When the igniter **50** is activated, the combustion of the igniter powder increases the pressure in the igniter **50**, which ruptures the tip of the squib cup **51** of the igniter **50**, causing flames to flow out of the tip of the squib cup **51** of the igniter **50** toward the gas generating agent **31** inside the housing **10**.

(27) The flame thus flowed ignites and burns the gas generating agent **31** in the housing **10** to generate a large amount of gas. By the combustion of the gas generating agent **31**, the pressure in the housing **10** increases, and the generated gas ruptures the weak member of the bottom surface portion **36a** of the bypass preventing member **36** and flows into the hollow portion **41a** and the internal space **12a**. Thereafter, the generated gas is ejected from the gas ejection port **11** to the outside of the gas generator **100** via the filter **41**, but since the generated gas passes through the filter **41**, the generated gas is cooled to a predetermined temperature. Then, the gas ejected from the gas ejection port **11** is guided into the airbag to inflate and deploy the airbag.

Main Features of Gas Generator **100**

(28) According to the present embodiment, since the holder is divided into three members (first holder **20**, second holder **25**, and third holder **53**), it is not necessary to form a complicated shape at a time, and thus it is possible to provide the gas generator **100** which is easier to manufacture than before.

(29) Furthermore, according to the present embodiment, since there is no empty volume around the igniter **50**, the pressure in the vicinity of the igniter **50** immediately after the activation of the igniter **50** can be increased as compared with a case where there is the empty volume. As a result, as compared with the case where there is the empty volume, the time (TTFG) from the start of activation until the generated gas is ejected from the gas ejection port can be shortened, and the combustibility of the gas generating agent **31** can be improved. That is, according to the present embodiment, by shortening the TTFG and improving the flammability of the gas generating agent **31**, the initial rise of the pressure in the housing **10**, which rises with the lapse of time, can be made sharper than before (improvement of P-t performance).

(30) Furthermore, according to the present embodiment, the third holder **53** can impart directivity in the direction of flame of the igniter **50**, and can reliably protect the circumferential wall **10d** deformed by the diameter reduction processing of the housing **10** from the pressure of the gas generated at the time of activation. In particular, since the third holder **53** has the surface **55** flush or substantially flush with the surface of the tip portion of the squib cup **51**, the third holder **53** can make the directivity of the direction of flame of the igniter **50** more accurate, and can more accurately protect the circumferential wall **10d** from the pressure of the gas generated at the time of activation.

(31) Furthermore, according to the present embodiment, the end portion of the first holder **20** on the side of the second holder **25** is sandwiched between the end portion of the second holder **25** on the side of the first holder **20** and the inner wall of the circumferential wall **10e** of the housing **10**, so that the first holder **20** can be more firmly fixed to the housing **10**. That is, even if the holder is divided into three members (first holder **20**, second holder **25**, and third holder **53**), sealability between the housing **10** and the holder (first holder **20**, second holder **25**, and third holder **53**) can be secured.

(32) Furthermore, according to the present embodiment, the third holder **53** can be used as a protective member of the caulking portion **24** of the second holder **25**.

(33) Furthermore, according to the present embodiment, since the first holder **20** and the third holder **53** are made of resin and the second holder **25** is made of metal or alloy, it is easier to mold

the holder than before, and the weight can be reduced as compared with a case where the entire holder is made of metal or alloy. That is, according to the present embodiment, it is possible to provide the gas generator **100** that can be more easily manufactured and lighter than the conventional gas generator and can reliably suppress the cost.

(34) Furthermore, in the present embodiment, since the closing member **12** and the housing **10** are caulked together, the caulking position is not particularly limited as long as it is a position where the closing member **12** can be fixed to the housing **10**. Therefore, since it is not necessary to manage the caulking position (accurate positioning or the like) as in the related art, the manufacturing process can be simplified.

(35) Furthermore, in the present embodiment, since the internal space **12a** is provided in the closing member **12**, the space volume can be increased as compared with the case of using the conventional closing member, and the internal pressure at the time of gas generation can be reduced.

Second Embodiment

(36) Next, a second embodiment of the present invention will be described with reference to FIG. 2. Note that, in the present embodiment, the parts having the same reference signs as those in the first embodiment up to the last two digits are the same as those in the first embodiment, and thus the description thereof may be omitted. Furthermore, in the present embodiment, portions that are not particularly described are similar to those of the first embodiment, and thus description and illustration thereof may be omitted.

(37) A gas generator **200** according to the present embodiment is different from the first embodiment mainly in that a first holder **120** is formed in a tubular shape so as to have a smaller maximum outer diameter than the first holder **20** of the first embodiment, and that a circumferential wall **110e** is formed following an outer shape of the first holder **120**.

(38) The first holder **120** is a substantially cylindrical member having a substantially constant outer diameter, and has a flange portion **126** formed at an end portion on a side of a second holder **125**. Note that the circumferential wall **110e** of a housing **110** in a portion corresponding to the flange portion **126** provided on the outer circumferential surface of the first holder **120** is radially reduced (caulked) to engage with the flange portion **126**, whereby the first holder **120** is caulked and fixed to the housing **110**. Furthermore, an end portion of the second holder **125** on a side of the first holder **120** is inserted into the first holder **120**. As a result, when the diameter of the circumferential wall **110e** of the housing **110** is reduced radially inward, the flange portion **126** is sandwiched between the end portion of the second holder **125** on the side of the first holder **120** and the inner wall of the circumferential wall **110e** of the housing **110**, and the first holder **120** can be more firmly fixed to the housing **110**. That is, sealability between the housing **110** and the holder can be secured. Here, as one modification, instead of the flange portion **126**, a simple end portion (having no flange protruding in the radial direction) may be provided, and the end portion may be sandwiched between the end portion of the second holder **125** on the side of the first holder **120** and the inner wall of the circumferential wall **110e** of the housing **110**.

(39) According to the present embodiment, the same effects as those of the first embodiment can be obtained. Furthermore, since the first holder **120** can be made smaller in size than the first holder **20** of the first embodiment, the gas generator **200** can be made lighter than the gas generator **100**. Furthermore, as compared with the first embodiment, the accuracy of the position of diameter reduction processing on the circumferential wall **110e** is not required.

Third Embodiment

(40) Next, a third embodiment of the present invention will be described with reference to FIG. 3. Note that, in the present embodiment, the parts having the same reference signs as those in the first embodiment up to the last two digits are the same as those in the first embodiment, and thus the description thereof may be omitted. Furthermore, in the present embodiment, portions that are not particularly described are similar to those of the first embodiment, and thus description and

illustration thereof may be omitted.

(41) A gas generator **300** according to the present embodiment is different from that of the first embodiment mainly in that a winding spring **235** is provided inside a bypass preventing member **236** and that a brim portion **256** is provided on a side of a closing member **212** of a third holder **253**.

(42) The winding spring **235** is disposed such that one end thereof is in contact with a bottom portion of the bypass preventing member **236**. Furthermore, the winding spring **235** is formed in a tubular shape, and presses a gas generating agent **231** toward an igniter **250** on the other end side.

(43) The brim portion **256** is a portion extending from the edge portion of the third holder **253** on a side of the closing member **212**, and is formed such that at least the tip portion of the brim portion **256** is deformed so as to be stuck to (come into contact with) an inner wall of a housing **210** by the pressure of the gas generated when the gas generating agent **231** burns. As a result, a circumferential wall **210d** of the housing **210** subjected to diameter reduction processing can be protected from the pressure of the gas. Therefore, the third holder **253** having the brim portion **256** can prevent the circumferential wall **210d** of the housing **210** subjected to the diameter reduction processing from being directly exposed to the heat of the generated gas, so that the strength of the circumferential wall **210d** can be maintained. Note that the tip portion of the brim portion **256** only needs to be in contact with the inner wall of the housing **210** at least at the time of activation, and may not be in contact with or may be in contact with the inner wall of the housing **210** in the initial state.

(44) Furthermore, in a case where it is assumed that there is no brim portion **256**, there is a space where the flame generated by the igniter **250** hardly reaches (near the corner portion between the vicinity of the edge portion of the third holder **53** and the inner wall of the housing **10** in the first embodiment). On the other hand, when the brim portion **256** is provided, the space is eliminated, so that the performance of the gas generator itself can be improved. In particular, by forming the brim portion **256** into a shape that allows the flame to easily reach (for example, a mortar shape), the performance of the gas generator itself can be further improved.

(45) Furthermore, since the brim portion **256** is integrally molded with the third holder **253**, the above-described effect by the brim portion **256** can be achieved without increasing the number of components.

(46) Furthermore, according to the present embodiment, it is possible to achieve the same effects as those of the first embodiment.

Fourth Embodiment

(47) Next, a fourth embodiment of the present invention will be described with reference to FIG. 4. Note that, in the present embodiment, the parts having the same reference signs as those in the first embodiment up to the last two digits are the same as those in the first embodiment, and thus the description thereof may be omitted. Furthermore, in the present embodiment, portions that are not particularly described are similar to those of the first embodiment, and thus description and illustration thereof may be omitted.

(48) A gas generator **400** according to the present embodiment is different from that of the first embodiment mainly in that a winding spring **335** is provided inside a bypass preventing member **336**, a brim portion **356** is provided on a side of a closing member **312** of a third holder **353**, the third holder **353** is not fixed to a housing **310**, the third holder **353** is fixed to an igniter **350**, and only a first holder **320** and a second holder **325** are sandwiched between a circumferential wall **310d** and a circumferential wall **310e** of the housing **310**.

(49) The winding spring **335** is disposed such that one end thereof is in contact with a bottom portion of the bypass preventing member **336**. Furthermore, the winding spring **335** is formed in a tubular shape, and presses a gas generating agent **331** toward the igniter **350** on the other end side.

(50) The third holder **353** has a brim portion **356** extending from an edge portion on a side of the closing member **312**, and a hole portion **357** (inside the third holder **353**) provided in a central

portion. The brim portion **356** can achieve the same effects as those of the brim portion **256** in the third embodiment. Note that the tip portion of the brim portion **356** may be in contact with the inner wall of the housing **310** at least at the time of activation similarly to the brim portion **256** in the third embodiment, and may not be in contact with or may be in contact with the inner wall of the housing **310** in the initial state.

(51) Furthermore, the hole portion **357** is press-fitted into a tip portion of a squib cup (cup-shaped member) **351** of the igniter **350**. Therefore, the third holder **353** is press-fitted and fixed to the igniter **350**. As a result, it is possible to prevent gas generated by combustion of the gas generating agent **331** from entering between the hole portion **357** and the squib cup **351** and to protect the circumferential wall **310d** of the housing **310** subjected to diameter reduction processing. That is, since the circumferential wall **310d** can be prevented from being exposed by the heat of the generated gas, the strength of the circumferential wall **310d** can be further maintained.

(52) Furthermore, according to the present embodiment, since it is not necessary to form portions such as the annular groove portions **54**, **154**, and **254** of the third holders **53**, **153**, and **253** of the first to third embodiments, complicated formation is not required and cost reduction is possible as compared with the first to third embodiments.

(53) Furthermore, according to the present embodiment, it is possible to achieve the same effects as those of the first embodiment.

(54) Although the embodiments of the present invention have been described above with reference to the drawings, it should be understood that the specific configuration is not limited to these embodiments. The scope of the present invention is defined not by the above description of the embodiment but by the claims, and further includes meanings equivalent to the claims and all modifications within the scope of the claims.

(55) For example, in each of the above embodiments, the flange portion (end portion) of the first holder is sandwiched between the end portion of the second holder on the side of the first holder and the inner wall of the circumferential wall of the housing subjected to diameter reduction processing, but the present invention is not limited thereto. That is, for example, at least a part of the outer circumference of the first holder and the end of the second holder on the side of the first holder and/or the inner wall of the housing may be bonded with an adhesive without sandwiching the flange (end) of the first holder. As another example, the first holder may be press-fitted into the housing.

(56) Furthermore, in each of the above embodiments, the caulking processing has been described as an example of the method of reducing the diameter of the housing, but any processing method capable of reducing the diameter of the housing may be used.

(57) Furthermore, although the outer diameter of the second holder in each of the above embodiments is larger than the outer diameter of the first holder on the other end side, the present invention is not limited thereto. For example, the outer diameter of the second holder may be the same as the outer diameter on the other end side of the first holder.

REFERENCE SIGNS LIST

(58) **10**, **110**, **210**, **310** housing **10a**, **10b**, **10c**, **10d**, **10e**, **110a**, **110b**, **110c**, **110d**, **110e**, **210a**, **210b**, **210c**, **210d**, **210e**, **210a**, **210b**, **210c**, **210d**, **210e** circumferential wall **11**, **111**, **211**, **311** gas ejection port **12**, **112**, **212**, **312** closing member **12a**, **112a**, **212a**, **312a** internal space **13**, **14**, **22**, **54**, **113**, **114**, **154**, **213**, **214**, **222**, **254**, **313**, **314**, **322** annular groove portion **20**, **120**, **220**, **320** first holder **21**, **23**, **121**, **123**, **221**, **223**, **321**, **323** fitting portion **24**, **124**, **224**, **324** caulking portion **25**, **125**, **225**, **325** second holder **26**, **126**, **226**, **326** flange portion **27**, **127**, **227**, **327** annular portion **31**, **131**, **231**, **331** gas generating agent **32**, **132** AI agent **35**, **135**, **235**, **335** winding spring **36**, **136**, **236**, **336** bypass preventing member **36a**, **136a**, **236a**, **336a** bottom surface portion **36b**, **136b**, **236b**, **336b** outer circumferential wall portion **41**, **141**, **241**, **341** filter **41a**, **141a**, **241a**, **341a** hollow portion **50**, **150**, **250**, **350** igniter **51**, **151**, **251**, **351** squib cup **52**, **152**, **252**, **352** terminal pin **53**, **153**, **253**, **353** third holder **55**, **155**, **255**, **355** surface **100**, **200**, **300**, **400** gas generator

Claims

1. A gas generator comprising: a housing being cylindrical and being loaded with a gas generating agent that generates a gas by burning, the housing including a filter inside through which the gas passes, and having a gas ejection port that ejects the gas formed at a position corresponding to the filter; an igniter capable of igniting and combusting the gas generating agent; a holder being tubular and holding a part of the igniter, the holder being fixed to a first end portion of the housing in an axial direction; and a closing member fixed to a second end portion of the housing in the axial direction, wherein the holder includes: a first holder being tubular and disposed on a side of the first end portion of the housing and capable of fitting an external connector that energizes the igniter on the side of the first end portion of the housing, a second holder being tubular and disposed on an inner side of the housing of the first holder in a state of being in contact with the first holder, an outer diameter of at least a contact portion, in contact with an inner wall of the housing, of the second holder is larger than an outer diameter of the first holder on a side of an end portion of the second holder or the outer diameter of at least the contact portion in contact with the inner wall of the housing of the second holder is same as the outer diameter of the first holder on the side of the end portion of the second holder, and a third holder being tubular and including, on an outer side surface, at least a small-diameter portion having a diameter smaller than a diameter of an end portion on a side opposite to a side of the second holder, and disposed on an inner side of the housing with respect to the second holder in a state of being in contact with the second holder, and a portion of the contact portion of the second holder corresponding to a portion on a side of the first holder and a portion corresponding to the small-diameter portion of the third holder are subjected to diameter reduction processing from an outer side of the housing, and the holder is fixed to the housing by sandwiching at least a portion of the first holder in contact with the second holder and a portion of the third holder in contact with the second holder together with the second holder with a portion of the housing subjected to the diameter reduction processing.
2. The gas generator according to claim 1, wherein at least a part on a side of the end portion of the second holder is fitted and inserted into the first holder, and an end portion of the first holder is fixed to the housing in a state of being sandwiched between the inner wall of the housing of a portion subjected to the diameter reduction processing at a position corresponding to an annular groove portion of the first holder and at least a part on a side of the end portion of the second holder.
3. The gas generator according to claim 1, wherein a brim portion is provided on a side of the closing member of the third holder, and at least at a time of activation, a tip portion of the brim portion is in contact with the inner wall of the housing.
4. A gas generator comprising: a housing being cylindrical and being that is loaded with a gas generating agent that generates a gas by burning, the housing including a filter inside through which the gas passes, and having a gas ejection port that ejects the gas formed at a position corresponding to the filter; an igniter capable of igniting and combusting the gas generating agent and including a tubular cup-shaped member on a side of the gas generating agent; a holder being tubular and holding a part of the igniter, the holder being fixed to a first end portion of the housing in an axial direction; and a closing member fixed to a second end portion of the housing in the axial direction, wherein the holder includes: a first holder being tubular and disposed on a side of the first end portion of the housing and capable of fitting an external connector that energizes the igniter on the side of the first end portion of the housing, a second holder being tubular and disposed on an inner side of the housing of the first holder in a state of being in contact with the first holder, an outer diameter of at least a contact portion, in contact with an inner wall of the housing, of the second holder is larger than an outer diameter of the first holder on a side of an end portion of the second holder or a diameter the outer diameter of at least the contact portion in

contact with the inner wall of the housing of the second holder is same as the outer diameter of the first holder on the side of the end portion of the second holder, and a third holder being tubular and including, on an outer side surface, at least a small-diameter portion having a diameter smaller than a diameter of an end portion on a side opposite to a side of the second holder, a brim portion being provided on a side of the closing member, the third holder being disposed on an inner side of the housing with respect to the second holder such that a tip portion of the brim portion comes into contact with the inner wall of the housing in a state of being in contact with the second holder and at least at a time of activation, a portion of the contact portion of the second holder corresponding to a portion on a side of the first holder and a portion corresponding to the small-diameter portion of the third holder are subjected to diameter reduction processing from an outer side of the housing, and the first holder and the second holder are fixed to the housing by sandwiching at least a portion of the first holder in contact with the second holder together with the second holder with a portion of the housing subjected to the diameter reduction processing, and the third holder is fixed to the igniter by press-fitting an inside of the third holder into the cup-shaped member of the igniter.

5. The gas generator according to claim 1, wherein a surface of the third holder on a side of the gas generating agent is formed such that the surface of the third holder on the side of the gas generating agent formed around a tip portion of the igniter is flush with a surface of the tip portion of the igniter.

6. The gas generator according to claim 1, wherein the second holder is a tubular member including a fitting portion that fits and holds the igniter inside, and a caulking portion that is provided at an opening end portion on a side of the end portion of the second holder and caulks and fixes the igniter, and the third holder is formed to cover the caulking portion.

7. The gas generator according to claim 1, wherein the first holder and the third holder are made of resin, and the second holder is made of metal or alloy.

8. The gas generator according to claim 2, wherein a brim portion is provided on a side of the closing member of the third holder, and at least at a time of activation, a tip portion of the brim portion is in contact with the inner wall of the housing.

9. The gas generator according to claim 2, wherein a surface of the third holder on a side of the gas generating agent is formed such that the surface of the third holder on the side of the gas generating agent formed around a tip portion of the igniter is flush with a surface of the tip portion of the igniter.

10. The gas generator according to claim 3, wherein a surface of the third holder on a side of the gas generating agent is formed such that the surface of the third holder on the side of the gas generating agent formed around a tip portion of the igniter is flush with a surface of the tip portion of the igniter.

11. The gas generator according to claim 4, wherein a surface of the third holder on a side of the gas generating agent is formed such that the surface of the third holder on the side of the gas generating agent formed around a tip portion of the igniter is flush with a surface of the tip portion of the igniter.

12. The gas generator according to claim 2, wherein the second holder is a tubular member including a fitting portion that fits and holds the igniter inside, and a caulking portion that is provided at an opening end portion on a side of the end portion of the second holder and caulks and fixes the igniter, and the third holder is formed to cover the caulking portion.

13. The gas generator according to claim 3, wherein the second holder is a tubular member including a fitting portion that fits and holds the igniter inside, and a caulking portion that is provided at an opening end portion on a side of the end portion of the second holder and caulks and fixes the igniter, and the third holder is formed to cover the caulking portion.

14. The gas generator according to claim 4, wherein the second holder is a tubular member including a fitting portion that fits and holds the igniter inside, and a caulking portion that is provided at an opening end portion on a side of the end portion of the second holder and caulks and

fixes the igniter, and the third holder is formed to cover the caulking portion.

15. The gas generator according to claim 5, wherein the second holder is a tubular member including a fitting portion that fits and holds the igniter inside, and a caulking portion that is provided at an opening end portion on a side of the end portion of the second holder and caulks and fixes the igniter, and the third holder is formed to cover the caulking portion.

16. The gas generator according to claim 2, wherein the first holder and the third holder are made of resin, and the second holder is made of metal or alloy.

17. The gas generator according to claim 3, wherein the first holder and the third holder are made of resin, and the second holder is made of metal or alloy.

18. The gas generator according to claim 4, wherein the first holder and the third holder are made of resin, and the second holder is made of metal or alloy.

19. The gas generator according to claim 5, wherein the first holder and the third holder are made of resin, and the second holder is made of metal or alloy.

20. The gas generator according to claim 6, wherein the first holder and the third holder are made of resin, and the second holder is made of metal or alloy.
